

Nitish Vikas Deshpande

PH.D. STUDENT IN WIRELESS COMMUNICATION WITH RESEARCH FOCUS ON MODELING, ANALYSIS, AND OPTIMIZATION OF RECONFIGURABLE ANTENNA ARRAYS.
EXPERIENCE IN 5G CELLULAR MODEM SYSTEMS FROM INTERNSHIPS AT QUALCOMM. RESEARCH EXPERIENCE ON 6G MASSIVE MIMO AT NOKIA BELL LABS.

UC San Diego · Electrical Engineering

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Education

University of California San Diego

San Diego, CA, USA

DOCTOR OF PHILOSOPHY (PH.D.), MAJOR IN ELECTRICAL ENGINEERING

Jan 2024 - June 2026 (Expected)

- Graduate student researcher with Prof. Robert W. Heath Jr.
- Graduate teaching assistant for the course Space-Time communication theory with Prof. Robert W. Heath Jr.
- Research focus: Reconfigurable antennas for MIMO wireless communication, circuit theory approach to communication

North Carolina State University

Raleigh, NC, USA

M.S., MAJOR IN ELECTRICAL ENGINEERING

August 2021 - Dec 2023

- G.P.A **4.0/4.0**
- Graduate Research Assistant with Prof. Robert W. Heath Jr.
- Relevant Coursework: RF Design for wireless, Advanced space-time communication, Antennas and arrays, Microwave design, LTE-5G.

Indian Institute of Technology, Kanpur

Kanpur, India

BACHELOR OF TECHNOLOGY, MAJOR IN ELECTRICAL ENGINEERING

July 2017 - May 2021

- C.G.P.A **9.4/10**
- Awarded Proficiency Prize for Best Project in Electrical Engineering

Work Experience

Modem systems intern, Qualcomm

Supervisors- Jae Won Yoo, Manish Jain, Abbas Termos

San Diego, CA, USA

AREA-EFFICIENT CHANNEL ESTIMATION FOR NEXT GENERATION CELLULAR MODEM

Jun. 2025 - Sept.2025

- Designed and validated a proof-of-concept to reduce computational complexity in DMRS-based channel estimation for PDSCH in next-generation 5G NR modems
- Analyzed throughput performance across diverse channel models, varying delay spreads, DMRS configuration types, and PRG sizes to optimize system-level tradeoffs

Modem systems intern, Qualcomm

Supervisors- Jae Won Yoo, Manish Jain, Aishwarya Kasa, Abbas Termos

San Diego, CA, USA

ENHANCED CHANNEL ESTIMATION WITH RX CORRELATION USING DMRS REFERENCE SIGNAL FOR 5G MODEM

Jun. 2024 - Sept.2024

- Proposed a low complexity algorithm leveraging Rx correlation to estimate the wireless channel
- Implemented the algorithm in Python in the end-to-end simulator and observed throughput gains for PDSCH

6G Radio systems research intern, Nokia Bell Labs

Supervisors- Saeed Khosravirad, Kumar Badheka, Jinfeng Du, Harish Viswanathan

Murray Hill, NJ, USA

PARASITIC RECONFIGURABLE ANTENNA ARRAY: A NEW PARADIGM OF ANALOG BEAMFORMING

Jun. 2023 - Aug.2023

- Proposed optimization methods for analog beamforming using parasitic reconfigurable antenna array
- Designed parasitic antenna in Feko software
- Received the "Outstanding student research award" for the intern project
- Submitted patent titled "Methods and devices for analog beamforming using parasitic resonating elements," to Finland Patent office

Communication systems research intern, Nokia Bell Labs

Supervisors- Saeed Khosravirad, Jinfeng Du, Harish Viswanathan

Murray Hill, NJ, USA

HARDWARE-AWARE ANALYSIS OF DENSE ARRAY MASSIVE MIMO

Jun. 2022 - Aug.2022

- Modeled wideband and dense array massive MIMO system using circuit theory
- Optimized array gain performance under several realistic hardware constraints
- Gained insights on the super-directivity phenomenon from a practical perspective

Research experience

Frequency-reconfigurable adhoc networks with dynamic metasurface antennas

UC San Diego, CA, USA

SUPERVISOR-**PROF. ROBERT W. HEATH JR.**, SPONSOR-**ARMY RESEARCH OFFICE**

Oct.2024-Present

- Proposed a new paradigm for adhoc networks that leverages the frequency-reconfigurability of dynamic metasurface antennas

Beamforming with hybrid reconfigurable parasitic antenna arrays

UC San Diego, CA, USA

SUPERVISOR-**PROF. ROBERT W. HEATH JR.**, SPONSOR-**NOKIA BELL LABS**

Aug.2023-Present

- Developing optimization methods for configuring parasitic antenna array using a multi-port circuit theory approach

Frequency-selective beamforming and beam training with dynamic metasurface antenna

UC San Diego, CA, USA

SUPERVISOR-**PROF. ROBERT W. HEATH JR.**, SPONSOR-**QUALCOMM**

Aug.2023-Present

- Developing optimization methods for configuring dynamic metasurface antennas for wideband communication
- The project won the “Qualcomm Innovation Fellowship” award for the academic year 2023-24

Achievable rate analysis with Bode-Fano wideband matching theory

NC State, Raleigh, NC, USA

SUPERVISOR-**PROF. ROBERT W. HEATH JR.**, SPONSOR-**NOKIA BELL LABS**

Jan.2023-May.2024

- Proposed a general bound on the MISO achievable rate over all linear and passive matching networks
- Developed a practical approach to approximate the achievable rate bound using ADS circuit design software

Analysis of near-field and wideband communication systems

NC State, Raleigh, NC, USA

SUPERVISOR-**PROF. ROBERT W. HEATH JR.**, SPONSOR-**NOKIA BELL LABS**

Jan.2022-June.2022

- Analyzed beamforming gain for a general near-field and wideband channel
- Proposed a frequency-selective distance metric to determine the far-field to near-field boundary for a wideband system
- Uncovered a fundamental tradeoff between the aperture and bandwidth

True-time-delay beamforming optimization for wideband systems

NC State, Raleigh, NC, USA

SUPERVISOR-**PROF. ROBERT W. HEATH JR.**, SPONSOR-**NOKIA BELL LABS**

Aug.2021-Dec.2021

- Proposed delay design for true-time-delay beamforming architectures for a general channel with multi-path and near-field effects
- Presented a poster on “Nonuniform true time delay precoding in wideband MISO systems,” in Asilomar conference on signals, systems, and computers in Pacific Grove, CA, in Nov 2022.

Teaching experience

Graduate teaching assistant for EE 287 in Spring 25 (Graduate class)

UC San Diego, CA, USA

INSTRUCTOR-**PROF. ROBERT W. HEATH JR.**, COURSE NAME-**SPACE-TIME COMMUNICATION THEORY**

March.2025-June.2025

- Grading weekly HWs on the Gradescope platform
- Conducting weekly office hours to assist students in HWs and understanding course material
- Guest lecture on topics like single user SISO information theory, waterfilling algorithm for capacity maximization, massive MIMO

Guest lecturer for ECE 301 in Fall 23 (Undergraduate class)

NC State, Raleigh, NC, USA

INSTRUCTOR-**PROF. MIGUEL RODRIGO CASTELLANOS**, COURSE NAME-**LINEAR SYSTEMS**

Oct. 2023-Nov. 2023

- Delivered lectures on topics spanning discrete time Fourier transform and examples of Fourier transform pairs - convolution and multiplication property, time and frequency shifting/ scaling, Parseval's theorem

Publications

JOURNAL AND MAGAZINE ARTICLES

[1] R. W. Heath, J. Carlson, N. V. Deshpande, M. R. Castellanos, M. Akrouf, C. B. Chae, “The tri-hybrid MIMO architecture,” accepted in *IEEE Communications Magazine* 2025

[2] N. V. Deshpande, M. R. Castellanos, S. R. Khosravirad, J. Du, H. Viswanathan, and R. W. Heath, “[Beamforming with hybrid reconfigurable parasitic antenna arrays](#),” accepted in *IEEE Transactions on Wireless Communications*, 2025.

[3] N. V. Deshpande, J. Carlson, M. R. Castellanos, and R. W. Heath, “[Frequency-selective beamforming and single-shot beam training with dynamic metasurface antennas](#),” under major revisions in *IEEE Transactions on Communications*, 2025.

[4] N. V. Deshpande, M. R. Castellanos, S. R. Khosravirad, J. Du, H. Viswanathan, and R. W. Heath, “[A generalization of the achievable rate of a MISO system using Bode-Fano wideband matching theory](#),” in *IEEE Transactions on Wireless Communications*, vol. 23, no. 10, pp. 13313-13329, Oct 2024.

[5] N. V. Deshpande, M. R. Castellanos, S. R. Khosravirad, J. Du, H. Viswanathan, and R. W. Heath, “[A wideband generalization of the near-field region for extremely large phased-arrays](#),” in *IEEE Wireless Commun. Lett*, 2022.

[6] N. V. Deshpande, S. Dey, D. Amudala, and R. Budhiraja, “[Spatially-Correlated IRS-Aided Multi-User FD mMIMO Systems: Analysis And Optimization](#),” in *IEEE Transactions on Communications*, vol. 70, no. 6, pp. 3879-3896, June 2022.

CONFERENCE PROCEEDINGS

[1] N. V. Deshpande, I. Kilinc, J. Carlson, and R. W. Heath, “Frequency-reconfigurable adhoc networks with dynamic metasurface antennas,” accepted in *IEEE Military Communications Conference 2025*, Los Angeles, CA, USA.

[2] N. V. Deshpande, J. Carlson, M. R. Castellanos, and R. W. Heath, “Frequency-selective beamforming with dynamic metasurface antennas,” accepted in *2025 IEEE International Conference on Communication*, Montreal, Canada 2025.

[3] N. V. Deshpande, M. R. Castellanos, S. R. Khosravirad, J. Du, H. Viswanathan, and R. W. Heath, “[Achievable rate of a SISO system under wideband matching network constraints](#),” in *GLOBECOM 2023 - 2023 IEEE Global Communications Conference*, Kuala Lumpur, Malaysia, 2023, pp. 7514-7519.

[4] N. V. Deshpande, S. R. Khosravirad, J. Du, H. Viswanathan, M. R. Castellanos, R. W. Heath, “[Analysis of dense array massive MIMO with practical constraints](#),” in 26th Workshop Smart Antennas, Feb. 2023.

[5] N. V. Deshpande, M. R. Castellanos, R. W. Heath, “[Nonuniform true time delay precoding in wideband MISO systems](#),” in Proc. Asilomar Conf. Signals, Syst., Comput., Nov. 2022. Conference presentation available [here](#).

[6] N. V. Deshpande, S. Dey, D. Amudala, E. Sharma and R. Budhiraja, “[Analysis of Statistical CSI-based Optimized Phase-Shift IRS-aided FD mMIMO System](#),” in *2021 IEEE Global Communications Conference*, Dec 2021. Conference presentation available [here](#).

Patents

[1] N. Deshpande, D. Badheka, S. R. Khosravirad “Methods and devices for analog beamforming using parasitic resonating elements,” submitted to *Finland, National Patent, Application number: 20246078*

Graduate coursework

Signal Processing and Machine Learning for Advanced MIMO, RF Design for wireless, Electromagnetic fields, Advanced space-time communication, Analog circuits, Optimization and algorithms, Antennas and arrays, Advanced microwave design, LTE and 5G, Wireless communication lab with USRPs

Honors & Awards

2024-25	Shannon Graduate Fellowship , Outstanding research in information theory applications	UC San Diego
2024	Brooklyn 6G Summit panelist , Participated in the 6G Graduate students panel moderated by Peter Vetter and shared my research experience in grad school	Nokia Bell Labs/ NYU
2023-24	Qualcomm Innvation Fellowship , Awarded for project proposal on “Dynamic metasurface antennas for wideband,” in collaboration with Joseph Carlson	Qualcomm/ UC San Diego
2023	Outstanding student research award , Summer intern project on “Parasitic reconfigurable antenna array,” received best intern project award in radio systems research at Nokia Bell Labs	Nokia Bell Labs, Murray Hill, NJ
2022	College of Engineering Travel grant , Awarded for Asilomar conference travel to Pacific Grove, CA	NC State
2021	Proficiency Prize , Project on intelligent reflecting surface received best UG project in EE	IIT Kanpur

Skills

Languages Python, MATLAB, C/C++, Verilog, VHDL

Software AWR, Feko, HFSS, ADS, Cadence, Xilinx ISE, Arduino IDE, LabView, Proteus, LaTeX

Hardware FPGA, USRP, micro-controllers boards like arduino/NodeMCU, Single Board Computers like Rpi, IOT boards like ESP